Printed Pages:02 **Sub Code:RME 702**

Paper Id: 231968 **Roll No.** | | | | | | | | | | |

B. TECH
(SEM VII) THEORY EXAMINATION 2022-23
AUTOMOBILE ENGINEERING

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2x7 = 14**

(a) Give any four classifications of automobile.
(b) Describe the term "firing order"use in IC engine.
(c) What is the need of clutch in transmission system?
(d) What do you understand by the term "Steering ratio"?
(e) What do you understand by the term "Breaking ratio"?
(f) What is the use of carburetor?
(g) Describe Fuel additives.

SECTION B

2. Attempt any *three* of the following: **7x3 = 21**

(a) Briefly explain the variation of resistance to the vehicle motion at the various speeds and total resistance to the vehicle motion.
(b) What is the function of a clutch? Discuss various factors affecting the torque transmission in a clutch.
(c) Describe the air braking system with a neat sketch. State the advantages of it.
(d) Write short notes on Hybrid vehicle and its types.
(e) Explain the properties of Fuel additives in brief.

SECTION C

3. Attempt any *one* part of the following: **7x1 = 7**

(a) Describe different types of Pistons. How does the 2-stroke piston differ from 4-stroke piston of a vehicle.
(b) What is the necessity of gear box? Explain the working of a sliding mesh gear box.

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4. Attempt any *one* part of the following: 7x1 = 7

(a) Explain the working &Draw a labelled diagram of a single plate clutch.
(b) What is overdrive? Explain its construction and discuss its working briefly.

5. Attempt any *one* part of the following: 7x1 = 7

(a) Explain telescopic shock absorber with diagram.
(b) Explain transfer of weight during application. Discuss how it affect wheel skidding.

6. Attempt any *one* part of the following: 7x1 = 7

(a) Write the working steps of Magneto Ignition System with diagram.
(b) Describe in detail Multipoint Fuel injection system discussing the main system components.

7. Attempt any *one* part of the following: 7x1 = 7

(a) Explain the role of Maintenance in automobile. Explain different types of maintenance with suitable example.
(b) In detail discuss different types of Alternative fuel used in automobile.

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